BRSM Final Project Report

Team - P < 1, Github -
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1 Introduction

This project provides a comprehensive analysis of how sentence memorability is influenced by the memorability of its constituent nouns and its syntactic voice. Using a continuous recognition memory paradigm, we investigate four primary hypotheses. First, we examine how noun-memorability composition (HH, HL, LH, LL) affects recognition accuracy. Second, we test whether syntactic voice (Active vs. Passive) modulates memorability. Extending beyond our initial mid-report analysis, we now also investigate how performance changes over time by testing for fatigue or learning effects across the three experimental blocks. Finally, we conduct an exploratory analysis of reaction times to determine if recognition speed differs across the noun-memorability conditions. By applying a sequence of non-parametric statistical tests, this study aims to provide a robust quantitative account of the cognitive factors that shape memory for language.

2 Mid-Report Recap

2.1 Domain Background

Human memory, particularly sentence memory, involves both the storage of meaning and the reconstruction of linguistic structure during recall. Early models proposed that sentences are stored as core semantic representations ("kernels") along with grammatical tags, suggesting that memory for syntax operates in an all-or-none manner.

However, Bregman and Strasberg (1968) challenged this view by showing that even when participants fail to recall exact syntax, their responses still preserve meaningful structural information. This indicates that sentences are reconstructed from partial memory traces rather than retrieved as complete units.

Further evidence from Begg (1971) demonstrated that memory for sentence meaning and exact wording are largely independent, with meaning often stored as conceptual or imagery-based representations.

Additionally, James, Thompson, and Baldwin (1973) showed that recall is influenced by linguistic biases such as noun saliency, where a prominent noun serves as the central theme. Due to a cognitive preference for active voice, sentences are more easily recalled when the salient noun appears in the subject position.

Overall, these studies suggest that sentence memorability arises from the interaction of semantic representation, linguistic structure, and cognitive reconstruction processes.

2.2 Data Overview and Description

The dataset consists of anonymized log files from 114 students who participated in a sentence recognition memory experiment as part of a course project. Each file contains trial-level data including stimulus details, event types, response accuracy, and reaction times across different conditions. During preprocessing, low-quality blocks were filtered out based on validation criteria, and one participant was removed entirely due to failing these checks. After cleaning, 73,503 valid event records were retained for analysis.

2.3 Hypothesis Testing (Partial)

Our mid-report analysis established the foundational findings of this study. The primary dependent variable was the **corrected memorability score** (Hit Rate - False Alarm Rate), which adjusts for individual response bias. Shapiro-Wilk tests confirmed that the data was not normally distributed ($p < .001$ for all conditions), mandating the use of non-parametric statistical methods.

The key findings were:

1. **Noun Memorability Effect (H1):** A Kruskal-Wallis test revealed a significant difference in corrected memorability scores across the four noun-memorability conditions ($H(3) = 10.00, p = .019$). Post-hoc Dunn's tests with Bonferroni correction showed that the LL (Low-Low) condition was significantly less memorable than the HH (High-High) and HL (High-Low) conditions. This supported the hypothesis that noun memorability impacts sentence recognition, suggesting a threshold effect where at least one high-memorability noun is sufficient to boost recall.

2. **Voice Effect (H2):** A Wilcoxon Signed-Rank test on paired data found no significant difference in corrected scores between Active and Passive sentences ($W = 1612.0, p = .200$). This indicated that, for this task, syntactic voice at test did not play a significant role in sentence memorability.

These initial results demonstrated that lexical properties (noun memorability) were a significant predictor of sentence recognition, while a key syntactic property (voice) was not. The final report builds on this by exploring temporal effects and response latencies.

3 Method

This study employed a series of statistical analyses to test four primary hypotheses regarding sentence memorability. A crucial preliminary step for all inferential tests was assessing the data for normality using the Shapiro-Wilk test and Q-Q plots. As the dependent variables consistently violated normality assumptions, a suite of non-parametric tests was used for all primary analyses. For multiple pairwise comparisons, p-values were adjusted using the Holm-Bonferroni method to control the family-wise error rate (FWER).

3.1 Hypothesis 1: Noun Memorability Composition

Normality of corrected memorability scores (DV) was assessed using the Shapiro-Wilk test for each condition (IV). Since the data violated normality, a Kruskal-Wallis H-test was performed as a non-parametric omnibus test analogous to a one-way ANOVA for independent groups. Following a significant result, Dunn's post-hoc test with Holm correction was used to conduct all six pairwise comparisons and identify which specific conditions differed.

3.2 Hypothesis 2: Syntactic Voice

The effect of syntactic voice (Active vs. Passive) on corrected memorability scores was assessed using a paired-sample design, as each participant contributed scores for both voices. After confirming the non-normality of the paired differences with a Shapiro-Wilk test, a Wilcoxon Signed-Rank test was employed. This test is the non-parametric equivalent of a paired t-test. As an exploratory follow-up, a potential Condition $\times$ Voice interaction was investigated by treating the data as 8 independent groups (e.g., HH-Active, HH-Passive, etc.) and performing a Kruskal-Wallis H-test to determine if there were any significant differences across these combined groups.

3.3 Hypothesis 3: Temporal Effects (Learning/Fatigue)

Hit rate per participant per block was used as the dependent variable, rather than the corrected memorability score, because the false alarm rate is a global, participant-level measure and cannot be decomposed by block. Both hit rate and false alarm rate can decrease due to fatigue, so corrected scores may not clearly reflect performance changes over time, hit rate provides a cleaner view of how recognition accuracy changes across blocks.

Normality of per-participant block hit rates was assessed using Shapiro-Wilk tests. Since all three blocks violated normality, a parametric repeated-measures ANOVA was ruled out. The analysis therefore used:

1. **Friedman test** - the non-parametric equivalent of a one-way repeated-measures ANOVA. It tests whether at least one block differs from the others, accounting for the within-subjects structure by ranking within each participant.

2. **Page's L trend test** - specifically designed for ordered alternatives in repeated measures. It tests the directional hypothesis Block 1 $\geq$ Block 2 $\geq$ Block 3 with greater power than Friedman when a monotone direction is predicted.

3. **Pairwise Wilcoxon Signed-Rank tests** (Holm-corrected) - post-hoc comparisons after a significant Friedman result. The signed-rank test uses within-participant paired differences, keeping the analysis consistent with the repeated-measures design.

Finally, to explore a potential Condition $\times$ Block interaction, separate Friedman tests were conducted on block-level hit rates within each of the four noun-memorability conditions.

3.4 Hypothesis 4: Reaction Time Analysis

Reaction times for correct "old" responses (hits) were analyzed at the trial level ($n \approx 1100$ trials per condition). Normality of RT distributions was assessed using Shapiro-Wilk tests. Since all conditions showed significant departures from normality, the analysis used:

1. **Kruskal-Wallis H-test** (omnibus, non-directional) - tests whether RT distributions differ across any pair of conditions without assuming normality. Trial-level data from different participants are treated as independent observations, which is appropriate here since different trials are genuinely independent.

2. **Dunn's post-hoc test** (Holm-corrected) - the appropriate follow-up to Kruskal-Wallis for independent groups, using the same pooled rank variance as the omnibus test. It identifies which specific condition pairs show significant differences through pairwise comparisons.

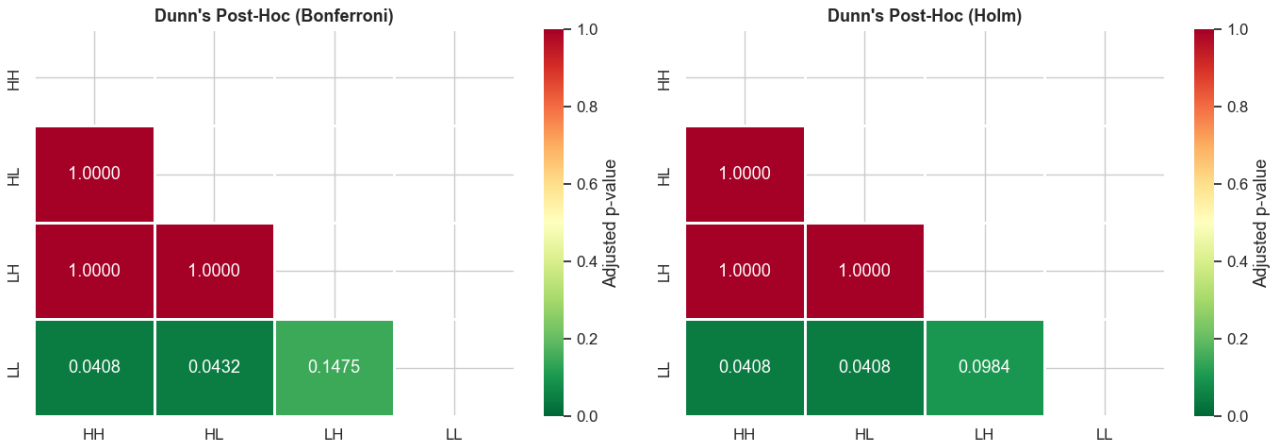

Figure 1: 4.1 p-values for pairwise comparisons across noun-memorability conditions.

3.5 Supplementary: Generalized Linear Model (GLM)

To complement the univariate tests and simultaneously model the influence of multiple predictors, a Gaussian Generalized Linear Model (GLM) was constructed. The model's outcome variable was the corrected memorability score, with subject noun memorability, object noun memorability, syntactic voice, and block number as predictors. Two models were fitted: a main-effects-only model (Model A) and a model including an interaction term between subject and object memorability (Model B). The models were compared using the Akaike Information Criterion (AIC), where the model with the lower AIC is considered to provide a better balance of fit and parsimony.

4 Results

4.1 Hypothesis 1: Noun Memorability Composition

Normality: Shapiro-Wilk tests confirmed non-normality in all conditions ($W < .94, p < .001$), necessitating non-parametric tests. Descriptive statistics (Table 1) reveal a dissociation between means and medians, suggesting a ceiling effect.

Condition	Mean	Median
HH	0.724	0.771
HL	0.725	0.750
LH	0.716	0.757
LL	0.666	0.688

Analysis: A Kruskal-Wallis test showed a significant difference across conditions, $H(3) = 10.00, p = .019, \eta^2_H = 0.016$. Dunn's post-hoc tests (Holm-corrected) confirmed that LL sentences were significantly less memorable than HH ($p = .041$) and HL sentences ($p = .041$). The LH vs. LL comparison was not significant ($p = .098$).

4.2 Hypothesis 2: Syntactic Voice

Normality: Shapiro-Wilk test on the within-participant differences (Active − Passive corrected score) indicated violation from normality ($W = 0.964$, $p = .004$), ruling out a paired t-test and directing the analysis to the Wilcoxon Signed-Rank test.

Descriptive Statistics:

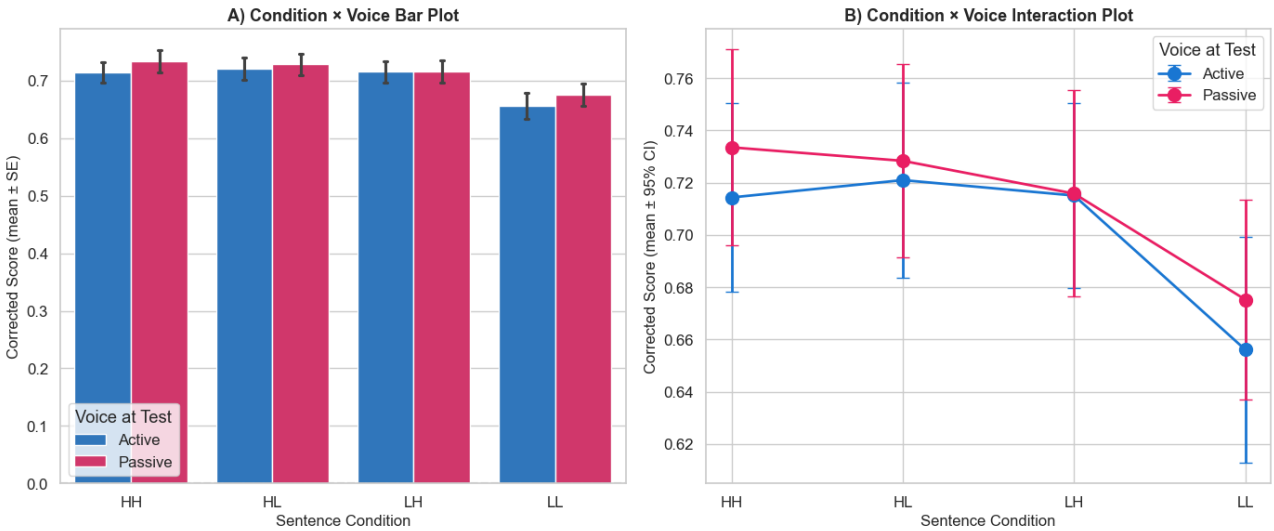

Figure 2: 4.2 Condition × Voice Corrected Memorability Score plots.

Voice	N	Mean	SD
Active	113	0.702	0.161
Passive	113	0.713	0.165

Passive sentences had a numerically higher mean corrected score, but the difference was negligible ($\Delta \approx 0.011$).

Wilcoxon Signed-Rank Test (paired): $W = 1612.0$, $p = .200$, $r = 0.158$ (small). The test was not significant. There is no evidence that syntactic voice at test presentation affected corrected memorability scores.

Exploratory: Condition × Voice (8-group Kruskal-Wallis): Shapiro-Wilk tests confirmed non-normality in all eight Condition × Voice groups (all $p < .001$). A Kruskal-Wallis test across the 8 groups yielded $H(7) = 14.03$, $p = .051$ - borderline non-significant. The descriptive pattern is shown below:

Condition	Active Mean	Passive Mean
HH	0.714	0.734
HL	0.721	0.728
LH	0.715	0.716
LL	0.656	0.675

A consistent Passive > Active trend is visible across all conditions, but the omnibus test did not reach significance and no post-hoc tests were conducted.

4.3 Hypothesis 3: Temporal Effects (Learning/Fatigue Across Blocks)

Normality: Shapiro-Wilk tests on per-participant hit rates confirmed non-normality in all three blocks (Block 1: $W = 0.853$, Block 2: $W = 0.804$, Block 3: $W = 0.862$, all $p < .001$).

Descriptive Statistics:

Block	Mean HR	Median HR	SD
Block 1	0.853	0.875	0.141
Block 2	0.833	0.875	0.185
Block 3	0.804	0.875	0.186

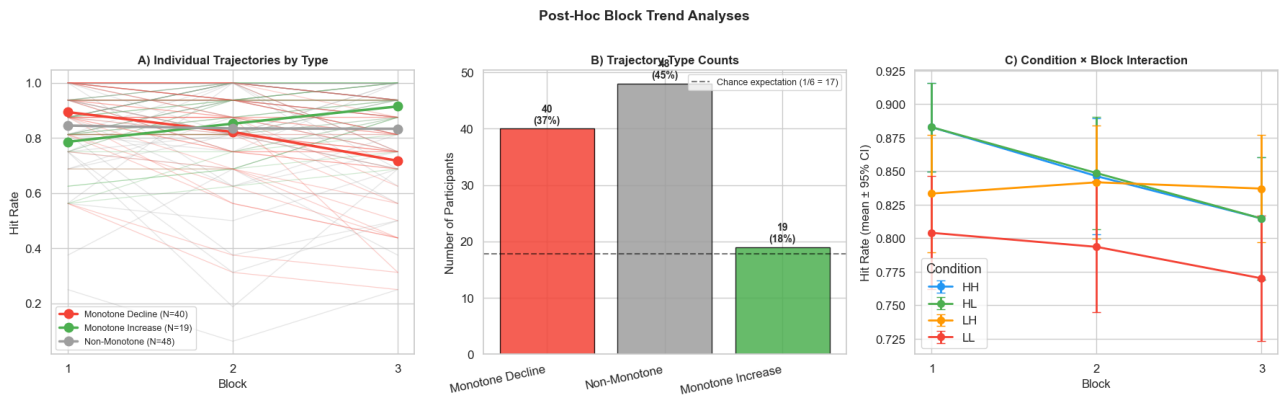

Figure 3: 4.3 Hit rate across blocks: (A) per-participant trajectories, (B) trajectory type counts, (C) Condition × Block interaction.

Mean hit rates declined monotonically (Δ Block 1→3 = −0.049). The median remained constant at 0.875 across all blocks, indicating that the decline was driven by a growing lower tail rather than a global shift.

Omnibus test (Friedman): $\chi^2(2) = 7.81$, $p = .020$. At least one block differed significantly.

Page's L trend test: $L = 1318.0$, $p = .010$, confirming a statistically significant monotone decrease Block 1 $\geq$ Block 2 $\geq$ Block 3.

Pairwise post-hoc tests (Wilcoxon, Holm-corrected):

Comparison	W	p_{Holm}	r	Significant?
Block 1 vs Block 2	1181.0	0.261	0.149	No
Block 1 vs Block 3	**1213.5**	**0.013**	**0.351**	**Yes**
Block 2 vs Block 3	1421.5	0.099	0.240	No (marginal)

Only the Block 1 vs Block 3 comparison survived correction ($p = .013$, $r = 0.351$, medium effect). The Block 1→Block 2 drop was not significant, suggesting the accuracy loss accumulates gradually and becomes detectable primarily at the end of the session.

Individual trajectory analysis: 37.4% of participants ($n = 40$) showed a monotone decline across blocks, significantly exceeding the chance expectation of 16.7%: $\chi^2(1) = 27.55$, $p < .001$.

Condition × Block interaction: Separate Friedman tests within each condition revealed that the block effect was significant only for HH sentences ($\chi^2(2) = 6.68$, $p = .036$), while LH sentences showed virtually no decline across blocks (0.829 → 0.843 → 0.836).

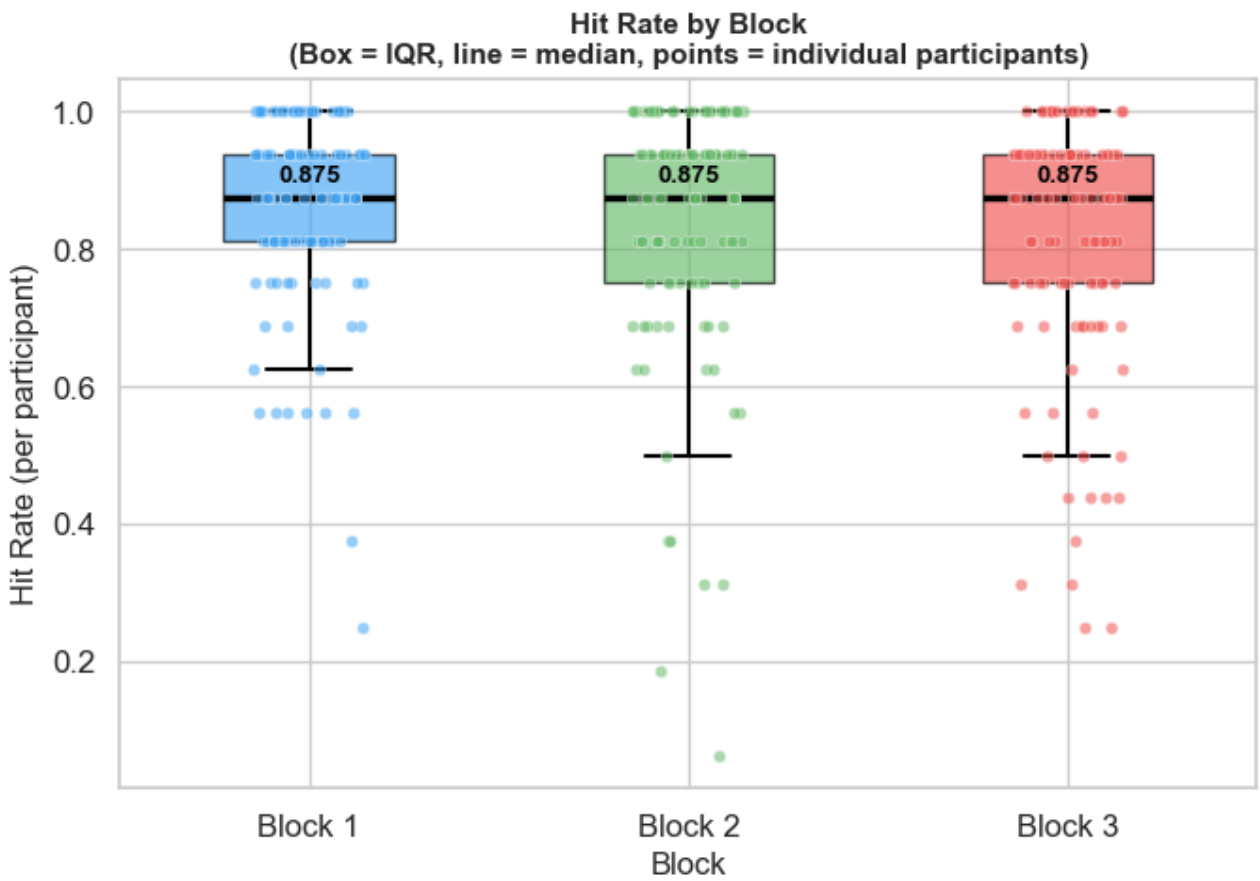

Figure 4: 4.3 Hit rates by block (box plot).

4.4 Hypothesis 4: Reaction Time Differences Across Conditions

Normality: Shapiro-Wilk tests on trial-level RT distributions confirmed non-normality in all four conditions (HH: $W = 0.930$, HL: $W = 0.942$, LH: $W = 0.929$, LL: $W = 0.936$, all $p < .001$).

Descriptive Statistics:

Condition	N	Mean (ms)	Median (ms)	SD (ms)
HH	1126	1725	1597	630
HL	1127	1712	1588	623
LH	1112	1661	1544	597
LL	1048	1761	1635	633

Empirical RT ranking (fastest → slowest): LH < HL < HH < LL. LH sentences were recognised fastest (median 1544 ms) and LL sentences slowest (median 1635 ms), a range of 91 ms across condition medians.

Overall test (Kruskal-Wallis): $H(3) = 14.87$, $p = .002$, $\eta^2_H = 0.003$. The test is statistically significant, however, the effect size is very small, with noun-memorability condition explaining less than 0.3% of the variance in RT.

Pairwise post-hoc tests (Dunn's, Holm-corrected):

Pair	p_{Holm}	r	Direction	Significant?
HH vs HL	0.788	−0.007	HH > HL	No
HH vs LH	0.118	−0.055	HH > LH	No
HH vs LL	0.213	+0.040	HH < LL	No
HL vs LH	0.184	−0.049	HL > LH	No
HL vs LL	0.184	+0.046	HL < LL	No
LH vs LL	**0.001**	**+0.096**	**LH < LL**	**Yes**

Only one pair survived Holm correction: LH vs LL ($p = .001$, $r = 0.096$, small effect). LH sentences were responded to significantly faster than LL sentences. All other pairwise comparisons were non-significant.

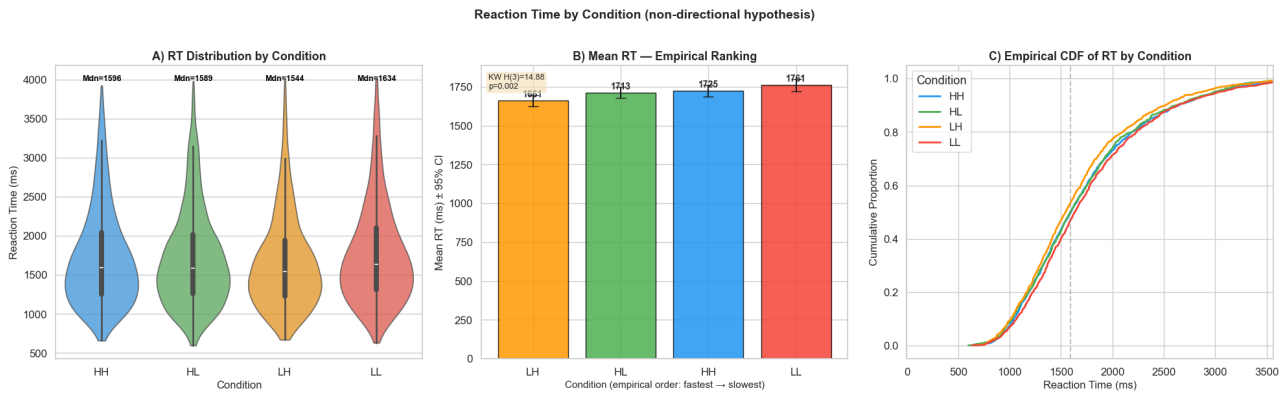

Figure 5: 4.4 Reaction time by condition: (A) violin plots with embedded box, (B) mean ± 95% CI sorted by empirical ranking, and (C) empirical CDF per condition.

4.5 Supplementary: Generalized Linear Model (GLM)

The GLM dataset comprised 2,656 observations (113 participants × 24 unique condition × voice × block cells). Two nested models were estimated.

Model A - Main Effects:

Predictor	β	SE	p	Sig.
Intercept	+2.074	0.120	<.001	***
Subject mem. [Low]	−0.251	0.074	.001	***
Object mem. [Low]	−0.170	0.074	.021	*
Voice [Passive]	+0.078	0.073	.287	
Block number	−0.149	0.045	.001	***
AIC = 3950.50				

Model B - Main Effects + Subject × Object Interaction:

Predictor	β	SE	p	Sig.
Intercept	+1.984	0.126	<.001	***
Subject mem. [Low]	−0.080	0.107	.455	
Object mem. [Low]	+0.006	0.108	.957	
Voice [Passive]	+0.078	0.073	.287	
Subject [Low] × Object [Low]	−0.325	0.148	.028	*
Block number	−0.149	0.045	.001	***
AIC = 3947.63				

Model comparison: Model B (AIC = 3947.63) is preferred over Model A (AIC = 3950.50), indicating that adding the Subject × Object interaction improves model fit without overfitting.

Interpretation of key effects:

- **Block number** ($\beta = -0.149$, $p = .001$): Recognition accuracy declined significantly with each successive block, consistent with the fatigue effect confirmed by H3.
- **Subject × Object interaction** ($\beta = -0.325$, $p = .028$): In Model B, the main effects of subject and object memorability become non-significant individually, but their interaction is significant. This indicates that the memorability penalty is not additive - the combination of a Low subject *and* a Low object (LL) produces a disproportionately larger drop in corrected score than either alone, consistent with the threshold effect described in H1.
- **Voice [Passive]** ($\beta = +0.078$, $p = .287$): Voice remained non-significant in both models, consistent with the H2 Wilcoxon result.

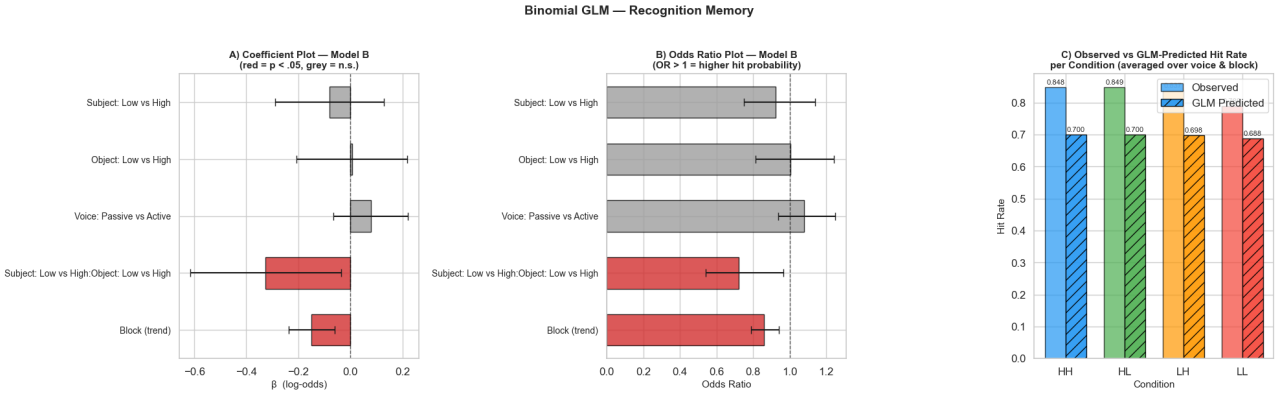

Figure 6: 4.5 GLM results: (A) coefficient forest plot for Model B, (B) observed vs. predicted corrected scores, (C) marginal predicted scores by condition.

5 Conclusion

5.1 H1: Noun Memorability Composition

Noun memorability significantly affected sentence recognition. The Kruskal-Wallis test showed a significant difference across the four conditions ($H(3) = 10.00$, $p = .019$). Post-hoc comparisons revealed that sentences with two low-memorability nouns (LL) were remembered significantly less often than sentences with at least one high-memorability noun (HH: $p = .041$, HL: $p = .041$). The LH condition was not significantly different from LL ($p = .098$), which may reflect a positional asymmetry: a vivid object noun is somewhat easier to overlook than a vivid subject noun when reconstructing a sentence from memory.

Key Takeaways

- Noun memorability has a clear and significant impact on sentence recognition.
- The corrected memorability scores followed the pattern: HL ≈ HH ≈ LH > LL.
- Having *at least one* high-memorability noun is enough to raise recognition performance to a near-ceiling level. Adding a second high-memorability noun does not produce a further measurable gain.

- The main performance drop occurs only when *both* nouns are low-memorability, suggesting a threshold rather than an additive effect.
- The divergence between mean and median scores points to a ceiling effect: many participants achieve near-perfect scores in the HH, HL, and LH conditions, compressing the upper end of the scale.

5.2 H2: Syntactic Voice

Syntactic voice did not significantly affect sentence recognition. The Wilcoxon Signed-Rank test on paired Active vs. Passive scores was not significant ($W = 1612.0$, $p = .200$, $r = 0.158$). An exploratory 8-group Kruskal-Wallis test across all Condition $\times$ Voice combinations also fell just short of significance ($H(7) = 14.03$, $p = .051$). While Passive sentences showed a slightly higher mean corrected score in every condition, the difference ($\Delta \approx 0.011$) was too small to reach statistical significance.

Key Takeaways

- The voice in which a sentence appears at test does not meaningfully change how well it is remembered.
- Active and Passive sentences produced nearly identical corrected scores (Active: $M = 0.702$, Passive: $M = 0.713$).
- The marginal $p = .051$ in the interaction test hints at a possible small Condition $\times$ Voice trend, but this cannot be treated as reliable evidence without a significant omnibus result.
- These findings are consistent with theories of sentence memory that emphasise semantic content over surface syntactic form: what a sentence *means* matters more than *how* it is phrased.

5.3 H3: Temporal Effects (Learning/Fatigue Across Blocks)

Recognition accuracy declined significantly over the course of the experiment. The Friedman test confirmed a significant block effect ($\chi^2(2) = 7.81$, $p = .020$) and Page's L trend test confirmed the decline was monotone ($L = 1318.0$, $p = .010$). The only significant pairwise contrast was Block 1 vs Block 3 ($W = 1213.5$, $p_{\text{Holm}} = .013$, $r = 0.351$), indicating a medium-sized cumulative fatigue effect by the end of the session.

Key Takeaways

- Performance declined steadily from Block 1 ($M = 0.853$) to Block 3 ($M = 0.804$), a drop of about 5 percentage points.
- The decline was gradual rather than sudden: the Block 1 to Block 2 drop was not significant on its own, but accumulated into a detectable effect by Block 3.
- 37.4% of participants showed a strict monotone decline across blocks - more than twice the 16.7% expected by chance ($\chi^2(1) = 27.55$, $p < .001$).
- The fatigue effect was not uniform across conditions. HH sentences showed the clearest block-to-block degradation ($\chi^2(2) = 6.68$, $p = .036$), while LH sentences were largely unaffected across blocks.
- Future studies using this paradigm should account for block position as a control variable, particularly when comparing conditions that appear in later blocks.

5.4 H4: Reaction Time Differences Across Conditions

Reaction times differed significantly across noun-memorability conditions ($H(3) = 14.87$, $p = .002$), but the effect was very small ($\eta^2_H = 0.003$). Only one pairwise comparison survived correction: LH sentences were recognised significantly faster than LL sentences ($p_{\text{Holm}} = .001$, $r = 0.096$). All other pairs were non-significant.

Key Takeaways

- The RT ranking (LH < HL < HH < LL) does not match the accuracy ranking from H1 (HH ≈ HL ≈ LH > LL). Speed and accuracy capture different aspects of the recognition process.
- LH sentences (low-memorability subject, high-memorability object) produced the fastest responses despite not producing the highest accuracy. A vivid object noun may act as a quick recognition cue, but the response is sometimes based on an incomplete or reconstructed memory trace.
- LL sentences produced both the lowest accuracy and the slowest responses, confirming that both memorability and retrieval speed are impaired when neither noun is distinctive.
- The very small effect size ($\eta^2_H = 0.003$) suggests that noun memorability has only a minor influence on *how fast* a sentence is recognised, even though it clearly affects *how accurately* it is recognised.

5.5 Overall Conclusion

This study examined how sentence memorability is shaped by the memorability of constituent nouns, syntactic voice, session fatigue, and reaction time, using a continuous recognition memory paradigm with 113 participants.

The central finding is that **noun memorability is the dominant lexical factor in sentence recognition**. Sentences containing at least one high-memorability noun were recognised significantly better than sentences containing two low-memorability nouns, and this held across both accuracy-based and model-based analyses. Crucially, the effect follows a threshold pattern rather than a simple additive one: a single high-memorability noun is sufficient to lift performance, and having two high-memorability nouns does not add further benefit. The GLM confirmed this picture - the Subject × Object interaction term was significant ($\beta = -0.325$, $p = .028$), showing that the LL penalty is disproportionately large rather than a simple sum of two independent noun-level penalties.

Syntactic voice had no significant effect on recognition accuracy. Active and Passive sentences were equally memorable, consistent with the idea that memory for sentences prioritises meaning over grammatical form. A marginal trend ($p = .051$) in the 8-group interaction analysis suggests the possibility of a small voice-by-condition interaction, but it falls short of the threshold for inference.

Fatigue is a real and measurable factor in this paradigm. Recognition accuracy fell significantly from the first to the third block, with a medium effect size ($r = 0.351$). This decline was systematic, not random, and was condition-specific: high-memorability sentences showed greater degradation over time, possibly because their initial advantage shrinks as cognitive resources are depleted.

Reaction time provides a complementary but distinct picture. The speed of recognition does not mirror accuracy: LH sentences were recognised fastest despite not being the most accurately recognised. This dissociation is consistent with a reconstruction account where

a salient object noun triggers a fast response that is occasionally based on an imperfect memory trace.

Together, these findings show that sentence memorability is a multi-factorial phenomenon shaped primarily by lexical content, with meaningful but secondary contributions from cognitive fatigue and, to a very limited extent, noun position effects in response speed.

References and Contribution

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- **James, C. T., Thompson, J. G., & Baldwin, J. M. (1973).** The reconstructive process in sentence memory. *Journal of Verbal Learning and Verbal Behavior*, 12(1), 51–63.
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Contribution

- **Arnav Deshmukh**: Analyzed group differences in noun memorability and implemented the Generalized Linear Model (GLM) for statistical inference. Worked for report development.
- **Nidhish Jain**: Conducted the literature review, formulated the Reaction Time (RT) hypotheses, studied the impact of block fatigue effects on participant performance and performed data transformation checks. Worked for report development.
- **Nilavra Ghosh**: Investigated the Voice Effect on memorability, developed data visualizations. Worked on developing GLM. Worked for report development.